

About the identification of tsunamis from surface measurements

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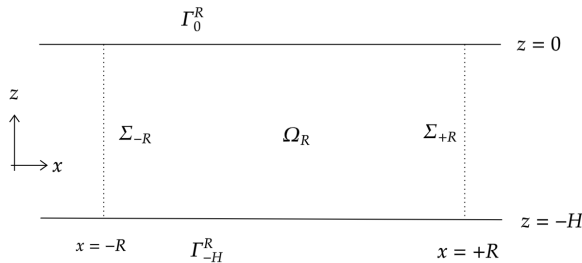
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Abstract

This talk deals with the reconstruction of the seabed deformations from the measurement of the level perturbation of the free ocean surface caused by a tsunami. A simple linear model obtained by linearisation of the Euler equations is used. We restrict ourselves to dimension 2 and to the frequency domain. Classically, such problems are solved in the literature using a Fourier transform along the horizontal direction, assuming a flat seabed. We have chosen to solve the inverse problem using a mixed variational Tikhonov formulation, the regularization parameter being chosen following the Morozov principle in the presence of noisy data. The Morozov value of the regularization parameter is obtained using a duality approach. This approach allows us to solve the problem without making any assumptions on the geometry of the seabed. An interesting particularity arises in the purely acoustic model when it comes to the lifting of noisy Neumann data: we propose a deterministic and a probabilistic method that we compare.

Keywords: Acoustic waves, Tikhonov regularization, Morozov choice, Duality

1 Setting of the inverse problem



Let us denote $\chi(x)$ the upward normal displacement of the sea bottom, c the celerity of acoustic waves and ω the frequency. The velocity potential $u(x, z)$ and χ are related to each other by the following forward problem: find $u \in H^1(\Omega_R)$

such that

$$\begin{cases} \Delta u + (\omega^2/c^2)u = 0 & \text{in } \Omega_R, \\ u = 0 & \text{on } \Gamma_0^R, \\ \partial_z u = -i\omega\chi & \text{on } \Gamma_{-H}^R, \\ \pm\partial_x u - T_{\pm}^a u = 0 & \text{on } \Sigma_{\pm R}. \end{cases}$$

The Dirichlet-To-Neumann operators

$T_{\pm}^a : H^{\frac{1}{2}}(\Sigma_{\pm R}) \rightarrow (H^{\frac{1}{2}}(\Sigma_{\pm R}))'$ are classically defined through a modal decomposition. From the measurements of the perturbation η of the free surface, we seek to reconstruct the displacement of the sea bottom χ . Thus, we seek $\chi \in L^2(\Gamma_{-H}^R)$ such that $\partial_z u = -i\omega\chi$, with $u \in H^1(\Omega_R)$ verifying:

$$\begin{cases} \Delta u + (\omega^2/c^2)u = 0 & \text{in } \Omega_R, \\ u = 0 & \text{on } \Gamma_0^R, \\ \partial_z u = -i\omega\eta & \text{on } \Gamma_{-H}^R, \\ \pm\partial_x u - T_{\pm}^a u = 0 & \text{on } \Sigma_{\pm R}. \end{cases} \quad (1)$$

We define $\mathcal{M} = \{\mu \in H^1(\Omega_R), \mu = 0 \text{ on } \Gamma_{-H}^R\}$ equipped with the norm $(\nabla \cdot, \nabla \cdot)_{L^2(\Omega_R)}$ and $H = \mathcal{M} \times L^2(\Gamma_0^R)$. The problem (1) is ill-posed and equivalent to the weak formulation: find $u \in H^1(\Omega_R)$ such that $u|_{\Gamma_0^R} = 0$ and $\forall \mu \in \mathcal{M}$

$$\begin{aligned} \int_{\Omega_R} \nabla u \cdot \nabla \bar{\mu} - \frac{\omega^2}{c^2} \int_{\Omega_R} u \bar{\mu} - \langle T_{+}^a u, \bar{\mu}|_{\Sigma_{+R}} \rangle \\ - \langle T_{-}^a u, \bar{\mu}|_{\Sigma_{-R}} \rangle = -i\omega \int_{\Gamma_0^R} \eta \bar{\mu}. \end{aligned}$$

In the following, we will write this weak formulation as $b(u, \mu) = \ell(\mu)$. Introducing the operator $B : H^1(\Omega_R) \rightarrow \mathcal{M}$ and $L \in \mathcal{M}$: $(Bu, \mu)_{\mathcal{M}} = b(u, \mu)$ and $(L, \mu)_{\mathcal{M}} = \ell(\mu)$, $\forall u \in H^1(\Omega_R), \forall \mu \in \mathcal{M}$, problem (1) is equivalent to find u such that:

$$Au = F \quad (2)$$

where $F = (L, 0) \in H$ and $A = (B, C) : H^1(\Omega_R) \rightarrow H$ and C being the trace operator on Γ_0^R .

2 Solving the inverse problem: a Tikhonov-Morozov approach

We consider noisy data F^δ with $\|F^\delta - F\|_H \leq \delta \leq \|F^\delta\|_H$. For $\epsilon > 0$, the standard Tikhonov

regularization consists in the well-posed weak formulation: find $u_\epsilon^\delta \in H^1(\Omega_R)$ such that $\forall v \in H^1(\Omega_R)$,

$$(Au_\epsilon^\delta, Av)_H + \epsilon(u_\epsilon^\delta, v)_{H^1(\Omega_R)} = (F^\delta, Av)_H. \quad (3)$$

We show that u_ϵ^δ is the solution of (3) if and only if $(u_\epsilon^\delta, \lambda_\epsilon^\delta := Bu_\epsilon^\delta - L^\delta)$ is the solution of

$$\begin{cases} \epsilon(u_\epsilon^\delta, v)_{H^1(\Omega_R)} + \overline{b(v, \lambda_\epsilon^\delta)} + (u_\epsilon^\delta, v)_{L^2(\Gamma_0^R)} = 0, \\ b(u_\epsilon^\delta, \mu) - (\lambda_\epsilon^\delta, \mu)_{\mathcal{M}} = \ell^\delta(\mu), \end{cases}$$

$\forall v \in H^1(\Omega_R), \mu \in \mathcal{M}$, with $\ell^\delta(\mu) = (L^\delta, \mu)_{\mathcal{M}}$.

In order to choose the regularization parameter ϵ , we introduce the Morozov Principle : there exists a unique $\epsilon > 0$ such that

$$\|Au_\epsilon^\delta - F^\delta\|_H = \delta.$$

Inspired by [1], one way to compute the choice of ϵ from the Morozov Principle is to solve the following minimization problem in H :

$$\inf_{q \in H} J^\delta(q) := \frac{1}{2} \|A^*q\|_{H^1(\Omega_R)}^2 + \delta \|q\|_H - \Re e \left\{ (F^\delta, q)_H \right\} \quad (4)$$

We denote by $u^\delta := u_{\epsilon(\delta)}^\delta$ the solution of (3) where $\epsilon(\delta)$ is the Morozov choice and q^δ the unique solution of (4). Then, one can show that

$$\epsilon(\delta) = \frac{\delta}{\|q^\delta\|_H}, \quad u^\delta = A^*q^\delta.$$

We remark that, to compute the value of $\epsilon(\delta)$, we only need to solve a constraints-free minimization problem, yet non-quadratic. Instead of minimizing J^δ , we introduce as in [2] a variational method based on a "quadratisation" of J^δ which avoids a gradient descent algorithm and nicely converges to the solution of (4).

3 Lifting the noisy Neumann data

To compute the Morozov solution u^δ , the amplitude of noise δ of the data F is required. In our purely acoustic case, δ coincides with the amplitude of noise on L , which is defined from the Neumann data η . Let us denote

$$\begin{cases} S : L^2(\Gamma_0^R) \rightarrow \mathcal{M} \\ f \mapsto L, \end{cases}$$

where L is the unique solution in $\mathcal{M}$ of the weak formulation:

$$\int_{\Omega_R} \nabla L \cdot \nabla \bar{\mu} \, dx dz = \int_{\Gamma_0^R} f \bar{\mu} \, dx, \quad \forall \mu \in M. \quad (5)$$

If d denotes the amplitude of noise on η , a simple way of estimating the amplitude noise δ on L is

$$\delta = \omega \|S\|d. \quad (6)$$

The estimate (6) is likely to overestimate δ . This is why, we propose an alternative way of computing δ based on a probabilistic model of the data. Assume that η is given on Γ_0^R by

$$\eta = \sum_{k=1}^N \eta_k \chi_k,$$

the χ_k being indicators functions of intervals of length $2R/N$ which do not overlap. The random variables η_k are independent and identically distributed variables following a Gaussian white noise. Instead of defining δ by (6), we can set

$$\delta = \omega r_N d, \quad r_N := \sqrt{\mathbb{E} \left[\frac{\|S\eta\|_{\mathcal{M}}^2}{\|\eta\|_{L^2(\Gamma_0^R)}^2} \right]}. \quad (7)$$

Such definition of δ is more realistic as it accounts for the fact that the measurements are discrete in practice. One can then show that

$$r_N = \sqrt{\frac{1}{2R} \sum_{k=1}^N \|S\chi_k\|_{\mathcal{M}}^2}.$$

To compute r_N we only need to solve N times the weak formulation (5) with $f = \chi_k$, $k = 1, \dots, N$. Some numerical experiments have shown the feasibility and the efficiency of our strategy to reconstruct the displacement χ .

References

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